MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION
FOURTH SEMESTER
Discipline Specific Elective (DSE) – MG4DSEMAT201
TRANSFORMS AND FOURIER SERIES
(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short Answer Questions

Answer any 5 questions
Each question carries 2 marks

1. Find the Laplace transform of t^2. [K] / [CO1]
2. State the First Shifting Theorem for Laplace transforms. [K] / [CO1]
3. State the Convolution Theorem for Laplace transforms. [K] / [CO2]
4. Find $L^{-1}\left\{\frac{1}{(s+2)(s+3)}\right\}$. [K] / [CO2]
5. Determine whether the function $f(x) = e^x$ is odd, even or neither odd nor even.[K] / [CO3]
6. Write the Fourier series of a 2π-periodic function and the expressions for a_0, a_n, b_n. [K] / [CO3]
7. Define the Fourier cosine transform of $f(x)$. [K] / [CO4]
8. Define the Fourier sine transform of $f(x)$. [K] / [CO4]

[2x5 = 10]

Part B
Short Essay Questions

Answer any 5 questions

Each question carries 6 marks

9. Using the first shifting theorem, find $L\{e^{2t}\sin(3t)\}$. [A] / [CO1]

10. Solve the IVP using Laplace transforms:

$$y'' + y = t, \qquad y(0) = 0, \qquad y'(0) = 1.$$

[A] / [CO1]

11. Using the convolution theorem, find

$$L^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\}.$$

[A] / [CO2]

12. Solve the integral equation using Laplace transforms:

$$y(t) = t + \int_0^t (t-\tau)y(\tau)\,d\tau.$$

[A] / [CO2]

13. Find the Fourier series expansion of $f(x) = x$ on $(-\pi, \pi)$. [A] / [CO3]

14. Check whether the following functions of period 2π, are even, odd or neither even nor odd.

    (a) $f(x) = x^2$, $(0 < x < 2\pi)$.

    (b) $f(x) = x^4$, $(0 < x < 2\pi)$.

[A] / [CO3]

15. Find the Fourier sine transform of $f(x) = e^{-ax}$, $a > 0$. [A] / [CO4]

16. Prove the Linearity of the Fourier transform. [A] / [CO4]

[6x5 = 30]

Part C
Essay Questions

Answer any 3 questions

Each question carries 10 marks

17. Solve using Laplace transforms:

$$y'' + y = u(t - \pi), \qquad y(0) = 0, \qquad y'(0) = 0,$$

where $u(t - a)$ is the unit step function. [A] / [CO1]

18. Solve using Laplace transforms:

$$y'' + 3y' + 2y = e^{-t}, \qquad y(0) = 1, \qquad y'(0) = 0.$$

[A] / [CO2]

19. Find the convolution $(f * g)(t)$ for $f(t) = t$ and $g(t) = \sin t$. Hence obtain

$$L^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\}.$$

[An] / [CO2]

20. (a) State whether the given function $f(x) = x^2/2$, $(-\pi < x < \pi)$ is even or odd. Find its Fourier series.

    (b) Then show that $1 + 1/4 + 1/9 + 1/16 + 1/25 + ... = \pi^2/6$

[An] / [CO3]

21. Find the Fourier transform of e^{-ax^2}, where $a > 0$. [A] / [CO4]

22. (a) Find the Fourier transform of $f(x) = k$ if $0 < x < a$ and $f(x) = 0$ otherwise.

    (b) Find the Fourier transform of xe^{-x^2}. [A] / [CO4]

[10x3 = 30]